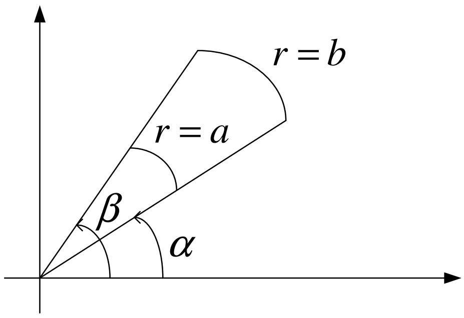
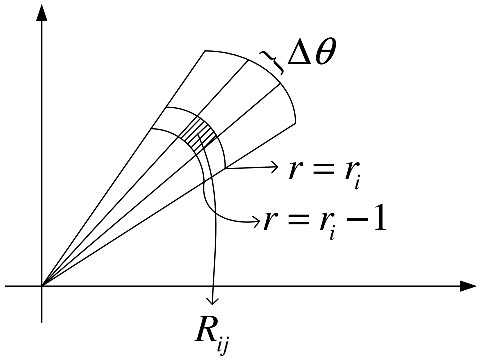
§15.4 Double Integrals in Polar Coordinates

Homework： 5,7,11,15,19,21,25,29,31,35

直角座標之積分變成極座標積分





$R=\{\ (r,\theta): a\le r\le b\ ,\ \alpha\le\theta\le\beta\ \}.$

$A_{ij}=\text{Area of }\ R_{ij}=\dfrac{1}{2}\left(r_i^2-r_{i-1}^2\right)\Delta\theta=\dfrac{1}{2}\left(r_i+r_{i-1}\right)\Delta r\Delta\theta$

當等分成“無窮”細時

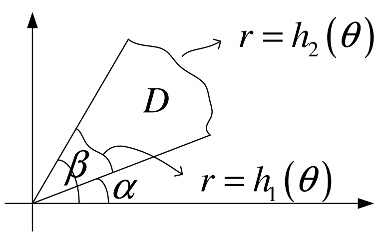
$A_{ij}\to dA=rdrd\theta$

Theorem：

(i) $\displaystyle\iint_R f(x,y)\,dA=\int_\alpha^\beta\int_a^b f(r\cos\theta,r\sin\theta)\ rdrd\theta$

(ii) $D=\{\ (r,\theta):\alpha\le\theta\le\beta\ ,\ h_1(\theta)\le r\le h_2(\theta)\ \}$

$$\Rightarrow\iint_D f(x,y)\,dA=\int_\alpha^\beta\int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta,r\sin\theta)\ rdrd\theta$$



Example 1： $\iint_D \left(3x+4y^2\right)\, dA.\quad D:1\le x^2+y^2\le 4\ ,\ y\ge 0.$

Solution：

$$\iint_D \left(3x+4y^2\right)\, dA$$
$$=\int_0^{\pi}\int_1^2 \left(3r\cos\theta+4r^2\sin^2\theta\right)\, rdrd\theta$$
$$=\frac{15\pi}{2}$$
$$\boxed{=\int_{-2}^{2}\int_{\sqrt{1-x^2}}^{\sqrt{4-x^2}} \left(3x+4y^2\right)\, dydx}$$

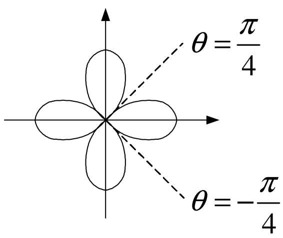
Example 2： Volume of $\begin{cases} z=1-x^2-y^2 \\ z=0 \end{cases}$

Solution：

$$\iint_D \left(1-x^2-y^2\right)\, dA \qquad D: x^2+y^2\le 1.$$
$$=\int_0^{2\pi}\int_0^1 \left(1-r^2\right) rdrd\theta=\frac{\pi}{2}$$

Example 3： Area of $r=\cos 2\theta$ (Area=Volume with height 1)

Solution：



$$\text{Area}=4\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\int_0^{\cos 2\theta}(1)\, rdrd\theta$$
$$=4\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\int_0^{\cos 2\theta} rdrd\theta$$
$$=\frac{\pi}{2}$$

Example 4： Volume of the solid that lies $\begin{cases} \text{under} \quad z=x^2+y^2 \\ \text{inside } x^2+y^2=2x \\ \text{above} \quad z=0 \end{cases}$

Solution：

$$V=\iint_D \left(x^2+y^2\right)\,dA$$
$$=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_0^{2\cos\theta} r^2 r\,dr\,d\theta$$
$$=\frac{3\pi}{2}$$

地基
$D:(x-1)^2+y^2=1$
$D: r=2\cos\theta$
屋頂
$z=x^2+y^2$
$z=r^2$

Example 5： Volume of $\begin{cases} x^2+y^2=4 \\ \dfrac{x^2}{16}+\dfrac{y^2}{16}+\dfrac{z^2}{64}=1 \end{cases}$

Solution：

$$V=2\int_0^{2\pi}\int_0^2 2\sqrt{16-r^2}\;r\,dr\,d\theta$$
$$=\frac{8\pi}{3}\left(64-24\sqrt{3}\right)$$

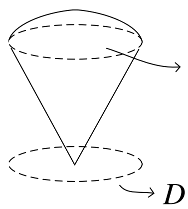
地基
$D: x^2+y^2=4$
$D: r=2$
屋頂
$z=\sqrt{64-4x^2-4y^2}$
$=\sqrt{64-4r^2}$
$=2\sqrt{16-r^2}$

Example 6： Volume of the solid that lies $\begin{cases} \text{below } x^2+y^2+z^2=1 \\ \text{above } z=\sqrt{x^2+y^2} \end{cases}$

$$\begin{cases} x^2+y^2+z^2=1 \\ z=\sqrt{x^2+y^2} \end{cases}$$

$$\Rightarrow\; x^2+y^2+\left(\sqrt{x^2+y^2}\right)^2=1$$

$$\Rightarrow\; x^2+y^2=\frac{1}{2}$$



$x^2+y^2=\dfrac{1}{2}$

$\left(z=\sqrt{\dfrac{1}{2}}\right)$

$D\quad x^2+y^2=\dfrac{1}{2}$

$(z=0)$

Solution：

$$V=\int_0^{2\pi}\int_0^{\frac{1}{\sqrt{2}}}\left(\sqrt{1-r^2}-r\right)\ rdrd\theta$$
$$=\frac{\pi}{3}\left(2-\sqrt{2}\right)$$

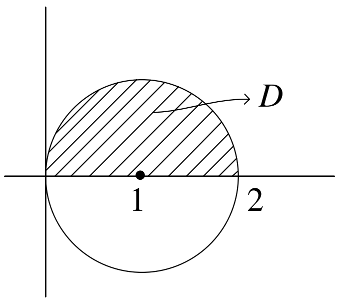
地基 $D: x^2+y^2=\frac{1}{2}$ $r=\frac{\sqrt{2}}{2}$
屋頂 $z=\sqrt{1-x^2-y^2}=\sqrt{1-r^2}$ $z=\sqrt{x^2+y^2}=r$

Example 7： Evaluate the iterated integral by converting to polar coordinates.

$$\int_0^2\int_0^{\sqrt{2x-x^2}}\sqrt{x^2+y^2}\ dydx$$

Solution：

$$y=\sqrt{2x-x^2}\Rightarrow x^2+y^2=2x\Rightarrow r=2\cos\theta$$



$$=\int_0^{\frac{\pi}{2}}\int_0^{2\cos\theta}r^2drd\theta$$
$$=\frac{16}{9}$$